

Jordan University of Science and Technology

Department of Mathematics and Statistics

Exam 1 (Math 103)-Fall 2024

الاسم: _____ الرقم الجامعي: _____ الشعبة: _____
 وقت المحاضرة: _____ اسم المدرس: الدكتور فنان عيسى
 الرقم المتسلسل: _____

Key solution

This exam consists of 15 multiple choice questions (2 points each). Put the correct answer in the table.

Question#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answers	B	A	C	D	B	D	A	C	A	C	D	D	B	C	A

1. If $f'(x) = \cos(x) - 2$, where $f\left(\frac{\pi}{2}\right) = 3$. Then

- A) $f(x) = \sin(x) - 2x$ (B) $f(x) = \sin(x) - 2x + \pi + 2$ C) $f(x) = \cos(x) + \pi + 2$ D) $f(x) = \sin(x) + \pi$

2. The integral $\int \frac{\cos(\sqrt{x})}{\sqrt{x}} dx =$

- (A) $2\sin(\sqrt{x}) + c$ B) $\sin(\sqrt{x}) + c$ C) $2\cos(\sqrt{x}) + c$ D) $\cos(\sqrt{x}) + c$

3. The value of the integral $\int_0^1 \tan^{-1}(x) dx =$

$$\tan\left(\frac{\pi}{4}\right) = \frac{\sin\left(\frac{\pi}{4}\right)}{\cos\left(\frac{\pi}{4}\right)} = \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} = 1$$

$$\tan^{-1}(1) = \frac{\pi}{4}$$

- A) $\frac{\pi}{2} - \frac{\ln(2)}{2}$ B) $\frac{\pi}{4} + \frac{\ln(2)}{2}$ (C) $\frac{\pi}{4} - \frac{\ln(2)}{2}$ D) $\frac{\pi}{4} + \frac{\ln(3)}{2}$

4. The value of the integral $\int_1^e \ln(x) dx =$

- A) $e-1$ B) -1 C) e (D) 1